

Towards the Shape Sensing of Multi-Segment Continuum Surgical Robots with Stretchable Resistive Strain Sensors

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INTRODUCTION

Shape sensing for continuum surgical robots remains an active and challenging research problem. A variety of different shape sensing modalities have been explored to date, including vision-based methods [1], electromagnetic tracking [2], and Fiber Bragg Grating (FBG) [3]. More recently, resistive strain sensors have started to gain attention as a viable and potentially simpler alternative [4,5].

In this paper, we explore the use of stretchable strain sensors [5] for distributed shape sensing in continuum surgical robots. These sensors operate analogously to conventional strain gauges, which measure changes in electrical resistance due to deformation, but over a substantially larger strain range. By installing multiple sensors along the robot's body, the proposed approach can capture the spatially varying curvature produced by external interactions, e.g., due to contact with an obstacle. In the following sections, we present the design, fabrication, and experimental evaluation of the proposed sensors. We describe their integration onto an 8 mm-diameter push-pull concentric-tube robot [6] and characterize their response under simulated obstacle-constrained bending. Experimental results demonstrate the feasibility of distributed sensing for capturing the bending of different robot sections.

MATERIALS AND METHODS

The stretchable strain sensors used in this study were introduced in recent work [5]. These sensors are fabricated starting from a thermoplastic polyurethane (TPU) film (ESTANE FS L75A4P, Lubrizol, USA), which is used as base layer. A gallium-indium eutectic (eGaIn) liquid metal gel [7] is stencil-printed onto the TPU substrate to form a 250 μm -wide sensing trace. A second TPU layer is hot-pressed at 130 $^{\circ}\text{C}$ for 5 minutes using a 1050 W press (TLM38385, Vevor, PRC) to encapsulate the liquid metal and form an airtight seal.

Given the intended application, it is natural to question whether eGaIn is suitable for use in a surgical device. Prior literature [8] indicates that both constituent metals are biologically safe, with Indium routinely used in dental fillings and Gallium recognized as a trace nutrient.

Two sensors were manufactured and attached to the body of a 3D-printed, 8 mm-diameter push-pull concentric tube robot (CPPR), as shown in Fig. 1. A hot-melt TPU

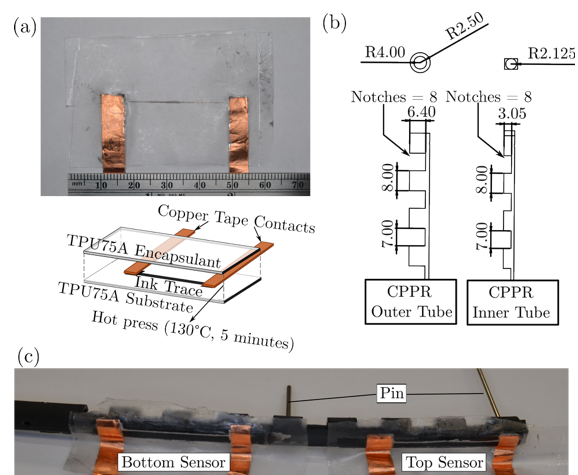


Fig. 1: Stretchable strain sensor fabrication and integration onto the body of a concentric push-pull robot (CPPR): (a) Fabricated sensor and corresponding schematic, (b) design parameters of the CPRR in millimeters, where R denotes radius, (c) CPRR with bottom and top sensors attached.

film (ESTANE FS HM70A71, Lubrizol) was used as an adhesive between the sensor and the robot body.

Robot bending experiments were performed to assess the sensors' ability to measure CPRR deflection. Bending was induced by translating the inner tube using a stepper motor-driven linear stage, with displacement measured via a magnetic encoder (AS5048A HTSP, ams OSRAM AG, Germany). Each actuation step applied a 5 mm translation over 20 seconds, followed by a 20 seconds hold. This protocol was repeated for five loading-unloading cycles. Sensor resistance was recorded through a Wheatstone bridge using an ADS1220 ADC (Texas Instruments). Translation and resistance signals were sampled at 100 Hz and timestamped in Python 3.8.1, and resistance data were smoothed in MATLAB using a moving average filter (window = 500).

During the tests, the bending angle of each robot segment was independently measured from camera images (Nikon D5100, Japan) using ImageJ, providing a ground-truth reference. Bending pins were affixed to the robot body (Fig. 1(c)) to facilitate measurement. To assess the sensors' ability to detect spatially varying curvature, a second set of experiments was performed in which a straight 3D-printed casing was installed on the lower half of the CPRR (Fig. 2), simulating an external obstacle that prevented bending in that section.

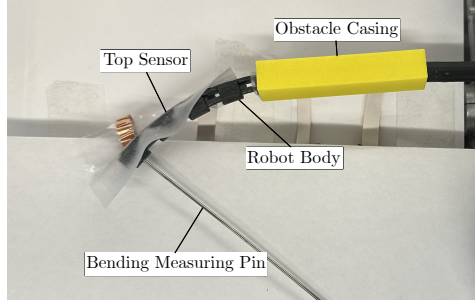


Fig. 2: 3D-printed casing to simulate the presence of an external obstacle over lower segment of the robot.

RESULTS

Figures 3(a) and 3(b) show raw sensor outputs alongside the corresponding ground-truth bending angles. Data from these experiments was used to calibrate a linear regression model to relate the observed change in resistance (ΔR) to the observed bending angles. The resulting fits achieved coefficients of determination of $R^2 = 0.982$ for the sensor mounted on the bottom half of the robot and $R^2 = 0.948$ for the sensor in the top half. These models were then applied to the raw sensor responses collected under casing-constrained conditions, as shown in Fig. 3(c), to predict the corresponding bending angles. The root-mean-square errors (RMSE) of these predictions were 2.03° for the lower segment and 4.91° for the upper segment.

DISCUSSION

Across all experiments, both sensors showed good agreement with ground truth measurements, indicating that resistance changes can be mapped to local curvature with good accuracy. These results suggests that arrays of similar sensors could be deployed along the robot body to recover its overall shape. Moreover, the current stencil-printing process supports further miniaturization, enabling higher sensor density and scalable integration through consolidated multi-channel routing at the robot base for finer robot sensing.

When the bottom section of the robot was constrained, the sensor in that section produced a small but nonnegligible negative response (Fig. 3(c)). This behavior is likely due to localized compression altering liquid metal distribution within the TPU films, affecting the measured resistance. Future designs incorporating mechanical isolation layers may help mitigate such compression-related artifacts.

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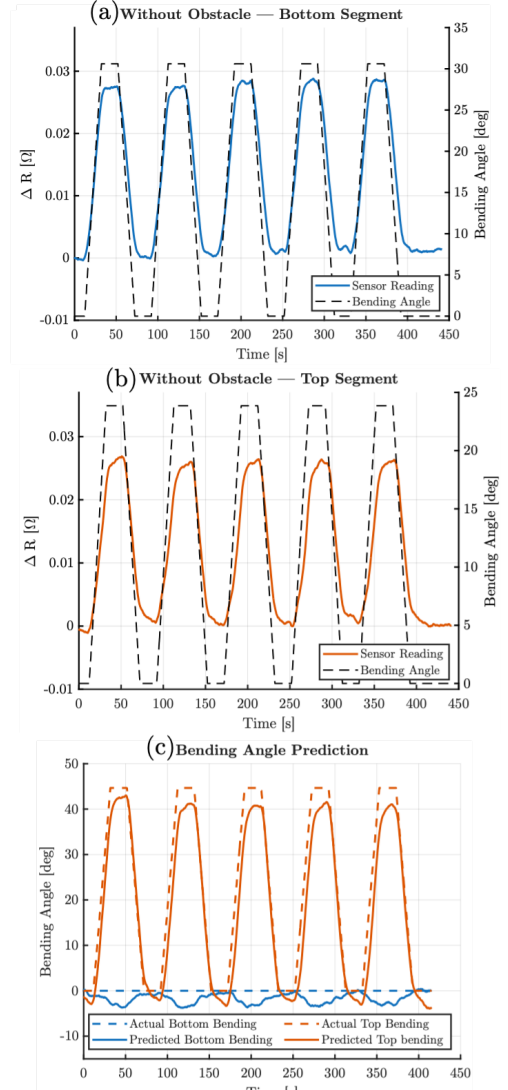


Fig. 3 Experimental results. (a) Raw output of the sensor mounted in the bottom section of the CPPR and corresponding ground-truth bending angle. (b) Same data for the sensor mounted in the top section of the CPPR. (c) Actual versus predicted bending angles during the constrained bending tests.

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